

Happy Paws Cat Adoption System

Project Title	Happy Paws Cat Adoption System
Student Name & Id	Sanjana Sayed Nuha (23-53087-3) Masrafi khan ornob (24-58728-2) Fairuz Dilshad (25-61786-1) Junaid Abdin Sakhir(24-56658-1)
Course / Section	Programming in python (D)
Instructor	Md. Tanzeem Rahat
Date of Submission	25/7/26

Introduction

The Happy Paws Cat Adoption System was created to simplify the cat adoption process through a user-friendly desktop application. It allows users to browse available cats, search using filters, and adopt a cat while keeping records updated automatically.

Objectives

- Develop a simple cat adoption system using Python.
- Apply Object-Oriented Programming concepts.
- Design an interactive GUI using Tkinter.
- Store and retrieve data using file handling.
- Use Python data structures such as lists, tuples, and sets.

Technologies Used

- Python
- Tkinter
- Text File (cats.txt)

OOP Concepts Used

The project uses a Cat class where each cat is represented as an object. It demonstrates classes, objects, a constructor (`__init__`), and encapsulation.

Data Structures Used

- List – Stores all cat objects.
- Tuple – Stores fixed breed and gender options.
- Set – Stores unique Cat IDs.

File Handling and Exception Handling

The program saves and loads data using cats.txt. It handles missing or inaccessible files using exception handling to improve reliability.

GUI Design

The interface includes buttons to view and search cats, filter options, a text area to display records, an entry field for Cat ID, an adoption button, and an exit button.

Program Features

- View available cats
- Search by breed, gender, and vaccination status
- Adopt a cat using its ID
- Automatically save updates
- Load saved data at startup

Testing

Tested features:

- View available cats
- Search
- Adoption
- Save and load data
- Exit application

GUI Screenshots



Happy Paws Cat Adoption

ID	Name	Breed	Age	Gender	Vaccinated	Status
C001	Luna	Persian	6 Months	Female	Yes	Available

View Available Cats

Breed

Persian

Gender

Female

Vaccinated

Yes

Search

Enter Cat ID To Adopt

Adopt Now

Exit

Happy Paws Cat Adoption

ID	Name	Breed	Age	Gender	Vaccinated	Status
C001	Luna	Persian	6 Months	Female	Yes	Available
C002	Leo	Tabby	1 Year	Male	Yes	Available
C003	Milo	Siamese	8 Months	Male	No	Available
C005	Mia	Mixed	5 Months	Female	No	Available
C006	Oliver	Persian	3 Years	Male	Yes	Available
C007	Lucy	Tabby	1 Year	Female	Yes	Available
C008	Bimba	Siamese	10 Months	Male	No	Available
C009	Lily	Ragdoll	2 Years	Female	Yes	Available
C010	Charlie	Mixed	7 Months	Male	Yes	Available

View Available Cats

Breed

Any

Gender

Any

Vaccinated

Any

Search

Enter Cat ID To Adopt

Adopt Now

Exit

Happy Paws Cat Adoption



ID	Name	Breed	Age	Gender	Vaccinated	Status
C002	Leo	Tabby	1 Year	Male	Yes	Available
C003	Milo	Siamese	6 Months	Male	No	Available
C005	Mais	Mixed	5 Months	Female	No	Available
C006	Oliver	Persian	3 Years	Male	Yes	Available
C007	Lucy	Tabby	1 Year	Female	Yes	Available
C008	Simba	Siamese	10 Months	Male	No	Available
C009	Lily	Ragdoll	2 Years	Female	Yes	Available
C010	Charlie	Mixed	7 Months	Male	Yes	Available

View Available Cats

Breed

Persian

Gender

Female

Vaccinated

Yes

Search

Enter Cat ID To Adopt

C00

Adopt Now

Exit